

INTEGRAL CALCULUS

Indefinite Integration

1. Let $f(x) = \int x^3 \sqrt{3-x^2} dx$. If $5f(\sqrt{2}) = -4$, then $f(1)$ is equal to :

माना $f(x) = \int x^3 \sqrt{3-x^2} dx$ है। यदि $5f(\sqrt{2}) = -4$ है, तो $f(1)$ बराबर है :

- (A) $-\frac{2\sqrt{2}}{5}$ (B) $-\frac{6\sqrt{2}}{5}$ (C) $-\frac{8\sqrt{2}}{5}$ (D) $-\frac{4\sqrt{2}}{5}$

Ans. B

Sol. $3 - x^2 = t^2$

$$x dx = -t dt$$

$$= \int (3 - t^2) t (-t) dt$$

$$= \int (t^4 - 3t^2) dt = \frac{t^5}{5} - t^3 + C$$

$$f(x) = \frac{(3-x^2)^{5/2}}{5} - (3-x^2)^{3/2} + C$$

$$f(\sqrt{2}) = -\frac{4}{5} \Rightarrow C = 0$$

$$f(1) = -\frac{6\sqrt{2}}{5}$$

2. If $\int e^x \left(\frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \frac{\sin^{-1} x}{(1-x^2)^{3/2}} + \frac{x}{1-x^2} \right) dx = g(x) + C$, where C is the constant of integration, then $g\left(\frac{1}{2}\right)$ equals :

यदि $\int e^x \left(\frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \frac{\sin^{-1} x}{(1-x^2)^{3/2}} + \frac{x}{1-x^2} \right) dx = g(x) + C$ है, जहाँ C समाकलन-अचर है, तो $g\left(\frac{1}{2}\right)$ बराबर है :

- (A) $\frac{\pi}{6} \sqrt{\frac{e}{2}}$ (B) $\frac{\pi}{6} \sqrt{\frac{e}{3}}$ (C) $\frac{\pi}{4} \sqrt{\frac{e}{2}}$ (D) $\frac{\pi}{4} \sqrt{\frac{e}{3}}$

Ans. B

Sol. $\int e^x \left(\underbrace{\frac{x \sin^{-1} x}{\sqrt{1-x^2}}}_{f(x)} + \underbrace{\frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} + \frac{x}{1-x^2}}_{f'(x)} \right) dx = g(x) + C$

$$\int e^x (f(x) + f'(x)) dx = e^x f(x) + C$$

So $g(x) = e^x f(x)$

$$g(x) = e^x \frac{x \sin^{-1} x}{\sqrt{1-x^2}}$$

$$g\left(\frac{1}{2}\right) = e^{\frac{1}{2}} \times \frac{1}{2} \frac{\sin^{-1}\left(\frac{1}{2}\right)}{\sqrt{1-\frac{1}{4}}} = \frac{\pi}{6} \sqrt{e}$$

3. Let $I(x) = \int \frac{dx}{(x-11)^{\frac{11}{13}} (x+15)^{\frac{15}{13}}}$. If $I(37) - I(24) = \frac{1}{4} \left(\frac{1}{b^{\frac{1}{13}}} - \frac{1}{c^{\frac{1}{13}}} \right)$, $b, c \in \mathbb{N}$, then $3(b+c)$ is equal to :

माना कि $I(x) = \int \frac{dx}{(x-11)^{\frac{11}{13}} (x+15)^{\frac{15}{13}}}$ है। यदि $I(37) - I(24) = \frac{1}{4} \left(\frac{1}{b^{\frac{1}{13}}} - \frac{1}{c^{\frac{1}{13}}} \right)$, $b, c \in \mathbb{N}$ है, तो $3(b+c)$ बराबर है :

(A) 40

(B) 39

(C) 26

(D) 22

Ans. B

Sol. $I(x) = \int \frac{dx}{(x-11)^{\frac{11}{13}} (x+15)^{\frac{15}{13}}}$

$$= \int \frac{dx}{\left(\frac{x-11}{x+15} \right)^{\frac{11}{13}} (x+15)^2}$$

$$\text{let } \frac{x-11}{x+15} = t$$

$$= \frac{1}{26} \int \frac{dx}{t^{\frac{11}{13}}}$$

$$\frac{26dx}{(x+15)^2} = dt$$

$$= \frac{1}{26} \left(\frac{t^{\frac{11}{13}+1}}{-\frac{11}{13}+1} \right) = \frac{1}{4} t^{\frac{2}{13}} = \frac{1}{4} \left(\frac{x-11}{x+15} \right)^{\frac{2}{13}}$$

$$I(37) - I(24) = \frac{1}{4} \left\{ \left(\frac{26}{52} \right)^{\frac{2}{13}} - \left(\frac{13}{39} \right)^{\frac{2}{13}} \right\}$$

$$= \frac{1}{4} \left\{ \frac{1}{4^{\frac{1}{13}}} - \frac{1}{9^{\frac{1}{13}}} \right\}$$

$$b = 4, c = 9$$

$$3(b + c) = 3(4 + 9) = 39$$

4. Let $\int x^3 \sin x dx = g(x) + C$, where C is the constant of integration. If $8 \left(g\left(\frac{\pi}{2}\right) + g'\left(\frac{\pi}{2}\right) \right) = \alpha\pi^3 + \beta\pi^2 + \gamma$, $\alpha, \beta, \gamma \in \mathbb{Z}$, then $\alpha + \beta - \gamma$ equals :

माना कि $\int x^3 \sin x dx = g(x) + C$ है, जहाँ C समाकलन-अचर है। यदि $8 \left(g\left(\frac{\pi}{2}\right) + g'\left(\frac{\pi}{2}\right) \right) = \alpha\pi^3 + \beta\pi^2 + \gamma$, $\alpha, \beta, \gamma \in \mathbb{Z}$ है, तो $\alpha + \beta - \gamma$ बराबर है :

(A) 48

(B) 55

(C) 62

(D) 47

Ans. B

Sol. $\int x^3 \sin x dx = g(x) + C$

$$\begin{array}{lcl} x^3 & + & \sin x \\ 3x^2 & - & -\cos x \\ 6x & + & -\sin x \\ 6 & - & \cos x \\ 0 & - & \sin x \end{array}$$

$$\therefore g(x) = -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$$

$$g\left(\frac{\pi}{2}\right) = \frac{3\pi^2}{4} - 6$$

$$\& g'\left(\frac{\pi}{2}\right) = \frac{\pi^3}{8}$$

$$8 \left[g\left(\frac{\pi}{2}\right) + g'\left(\frac{\pi}{2}\right) \right] = \pi^3 + 6\pi^2 - 48$$

$$\alpha + \beta - \gamma = 55$$

5. If $f(x) = \int \frac{1}{x^{\frac{1}{4}} \left(1 + x^{\frac{1}{4}}\right)} dx$, $f(0) = -6$, then $f(1)$ is equal to :

यदि $f(x) = \int \frac{1}{x^{\frac{1}{4}} \left(1 + x^{\frac{1}{4}}\right)} dx$, $f(0) = -6$, है, तो $f(1)$ बराबर है :

(A) $2 - \log_e 2$ (B) $4(\log_e 2 - 2)$ (C) $\log_e 2 + 2$ (D) $4(\log_e 2 + 2)$

Ans. B

Sol. $f(x) = \int \frac{1}{x^{\frac{1}{4}} \left(1 + x^{\frac{1}{4}}\right)} dx$

$$x^{\frac{1}{4}} = t$$

$$\left(\frac{1}{4}\right) x^{-\frac{3}{4}} dx = dt$$

$$f(x) = \int \frac{4 \cdot x^{\frac{3}{4}} dt}{t(1+t)}$$

$$\Rightarrow \int \frac{4t^3}{t(1+t)} dt$$

$$f(x) = 4 \int \frac{(t^2 - 1) + 1}{(1+t)} dt$$

$$f(x) = 4 \int \left[(t-1) + \frac{1}{(1+t)} \right] dt$$

$$f(x) = 4 \left(\frac{t^2}{2} - t + \ln(1+t) \right) + C$$

$$f(x) = 4 \left(\frac{x^{\frac{1}{2}}}{2} - x^{\frac{1}{4}} + \ln \left(1 + x^{\frac{1}{4}} \right) \right) + C$$

If $f(0) = -6$

$$f(0) = 4(\ln(1)) + C = -6$$

$$C = -6$$

$$f(x) = 4 \left(\frac{x^{\frac{1}{2}}}{2} - x^{\frac{1}{4}} + \ln \left(1 + x^{\frac{1}{4}} \right) \right) - 6$$

$$f(1) = 4 \left(\frac{1}{2} - 1 + \ln 2 \right) - 6$$

$$\Rightarrow 4 \left(\ln 2 - \frac{1}{2} \right) - 6$$

$$\Rightarrow 4 \ln 2 - 2 - 6$$

$$\Rightarrow (4 \ln 2 - 8) \Rightarrow 4(\ln 2 - 2)$$

6. If $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{1}{12} \tan^{-1}(3 \tan x) + \text{constant}$, then the maximum value of $a \sin x + b \cos x$, is :

यदि $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{1}{12} \tan^{-1}(3 \tan x) + \text{अचर है, तो } \sin x + b \cos x \text{ का उच्चतम मान है :}$

(A) $\sqrt{40}$

(B) $\sqrt{42}$

(C) $\sqrt{39}$

(D) $\sqrt{41}$

Ans. A

Sol. $\int \frac{\sec^2 x dx}{a^2 \tan^2 x + b^2}$

let $\tan x = t$

$\sec^2 x dx = dt$

$$\int \frac{dt}{a^2 t^2 + b^2}$$

$$\frac{1}{a^2} \int \frac{dt}{t^2 + \left(\frac{b}{a} \right)^2}$$

$$\frac{1}{a^2} \frac{1}{b} \tan^{-1} \left(\frac{t}{b} a \right) + c$$

$$\frac{1}{ab} \tan^{-1} \left(\frac{\alpha}{b} \tan x \right) + c$$

on comparing $\frac{a}{b} = 3$

$$ab = 12$$

$$a = 6, b = 2$$

maximum value of

$$6 \sin x + 2 \cos x \text{ is } \sqrt{40}$$

7. Let $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$. If $I(0) = 3$, then $I\left(\frac{\pi}{12}\right)$ is equal to :

माना $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$ है। यदि $I(0) = 3$, तब $I\left(\frac{\pi}{12}\right)$ बराबर होगा :

(A) $2\sqrt{3}$

(B) $\sqrt{3}$

(C) $3\sqrt{3}$

(D) $6\sqrt{3}$

Ans. C

Sol. $I(x) = \int \frac{6dx}{\sin^2 (1 - \cot x)^2}$ $I(0) = 3$

$$I\left(\frac{\pi}{12}\right) = ?$$

$$I(x) = \int \frac{6dx}{\sin^2 x \frac{(\sin x - \cos x)^2}{\sin^2 x}} = \int \frac{6dx}{\sin^2 x + \cos^2 x - 2 \sin x \cos x}$$

$$= \int \frac{6dx}{1 - 2 \sin x \cos x} = \int \frac{6 \sec^2 x dx}{1 + \tan^2 x - 2 \tan x}$$

$$= \int \frac{6 \sec^2 x dx}{(\tan x - 1)^2}$$

$$= \tan x = t$$

$$= \sec^2 x dx = dt$$

$$= \int \frac{6dt}{(t-1)^2} = 6 \int (t-1)^{-2} dt$$

$$I(x) = \frac{-6}{(t-1)} + C$$

$$I(x) = -\frac{6}{\tan x - 1} + C$$

$$I(0) = 3$$

$$3 = 6 + C$$

$$C = -3$$

$$I(x) = -\frac{6}{\tan x - 1} - 3$$

$$I\left(\frac{\pi}{12}\right) = -\frac{6}{2 - \sqrt{3} - 1} - 3$$

$$= -\frac{6}{1 - \sqrt{3}} - 3 = -\left(\frac{6 + 3 - 3\sqrt{3}}{1 - \sqrt{3}}\right)$$

$$= -\left(\frac{9 - 3\sqrt{3}}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}}\right)$$

$$= -\left(\frac{9 + 9\sqrt{3} - 3\sqrt{3} - 9}{1 - 3}\right) = \frac{6\sqrt{3}}{2} = 3\sqrt{3}$$

8. Let $\int \frac{2 - \tan x}{3 + \tan x} dx = \frac{1}{2}(\alpha x + \log_e |\beta \sin x + \gamma \cos x|) + C$, where C is the constant of integration. Then $\alpha + \frac{\gamma}{\beta}$ is equal to :

माना $\int \frac{2 - \tan x}{3 + \tan x} dx = \frac{1}{2}(\alpha x + \log_e |\beta \sin x + \gamma \cos x|) + C$ है, जहाँ C समाकलन अचर है। तो $\alpha + \frac{\gamma}{\beta}$ बराबर है :

(A) 3

(B) 1

(C) 7

(D) 7

Ans. D

Sol. $\int \frac{2 - \tan x}{3 + \tan x} dx = \int \frac{2 \cos x - \sin x}{3 \cos x + \sin x} dx$

Put, $2 \cos x - \sin x = a(-3 \sin x + \cos x) + b(3 \cos x + \sin x)$

$$a + 3b = 2 \quad \dots\dots\dots(i)$$

$$-3a + b = -1 \quad \dots\dots\dots(ii)$$

From equation (i) and (ii) $a = b = \frac{1}{2}$

$$I = \frac{1}{2} \int \frac{-3 \sin x + \cos x}{3 \cos x + \sin x} dx + \frac{1}{2} \int \frac{3 \cos x + \sin x}{3 \cos x + \sin x} dx$$

$$I = \frac{1}{2} \ln |3 \cos x + \sin x| + \frac{1}{2} x = \frac{1}{2} (\alpha x + \log |\beta \sin x + \gamma \cos x|)$$

On comparing, we get

$$\alpha = 1, \beta = 1, \gamma = 3$$

$$\alpha + \frac{\gamma}{\beta} = 1 + \frac{3}{1} = 4$$

9. The integral $\int \frac{(x^8 - x^2) dx}{(x^{12} + 3x^6 + 1) \tan^{-1} \left(x^3 + \frac{1}{x^3} \right)}$ is equal to :

समाकलन $\int \frac{(x^8 - x^2) dx}{(x^{12} + 3x^6 + 1) \tan^{-1} \left(x^3 + \frac{1}{x^3} \right)}$ बराबर है :

(A) $\log_e \left(\left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| \right)^{1/3} + C$

(B) $\log_e \left(\left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| \right)^{1/2} + C$

(C) $\log_e \left(\left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| \right) + C$

(D) $\log_e \left(\left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| \right)^3 + C$

Ans. A

Sol. $\int \frac{(x^8 - x^2) dx}{(x^{12} + 3x^6 + 1) \tan^{-1} \left(x^3 + \frac{1}{x^3} \right)}$

$$\tan^{-1} \left(x^3 + \frac{1}{x^3} \right) = t$$

$$\frac{\left(3x^2 - \frac{3}{x^4} \right) dx}{1 + \left(x^3 + \frac{1}{x^3} \right)^2} = dt$$

$$\frac{3(x^6 - 1)dx}{x^4 \left(1 + x^6 + \frac{1}{x^6} + 2 \right)} = dt$$

$$\frac{3(x^8 - x^2)dx}{(x^{12} + 3x^6 + 1)} = dt$$

$$= \frac{1}{3} \int \frac{dt}{t}$$

$$= \frac{1}{3} \ln |t| + c$$

$$\log_e \left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right|^{1/3} + c$$

10. For $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$, if $y(x) = \int \frac{\cos \sec x + \sin x}{\cos \sec x \sec x + \tan x \sin^2 x} dx$, and $\lim_{x \rightarrow \left(\frac{\pi}{2} \right)^-} y(x) = 0$ then $y\left(\frac{\pi}{4} \right)$ is equal to :

$x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$ के लिए यदि $y(x) = \int \frac{\cos \sec x + \sin x}{\cos \sec x \sec x + \tan x \sin^2 x} dx$ है तथा $\lim_{x \rightarrow \left(\frac{\pi}{2} \right)^-} y(x) = 0$ है, तो $y\left(\frac{\pi}{4} \right)$ बराबर है :

(A) $\tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$ (B) $\frac{1}{\sqrt{2}} \tan^{-1} \left(-\frac{1}{2} \right)$ (C) $-\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$ (D) $\frac{1}{2} \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$

Ans. B

Sol. $\int \left(\frac{1 + \sin^2 x}{\frac{1}{\cos x} + \frac{\sin^4 x}{\cos x}} \right) dx$

$$= \int \frac{(1 + \sin^2 x) \cos x}{1 + \sin^4 x} dx$$

$\sin x = t, \quad \cos x dx = dt$

$$\int \frac{1+t^2}{1+t^4} dt = \int \left(\frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}} \right) dt$$

$$= \int \frac{du}{u^2+2} \quad t - \frac{1}{t} = u$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \frac{u}{\sqrt{2}} + c \quad \left(1 + \frac{1}{t^2} \right) dt = dx$$

$$f(x) = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sin x - \frac{1}{\sin x}}{\sqrt{2}} \right) + c$$

$$f\left(\frac{\pi}{2}\right) = 0 \Rightarrow c = 0$$

$$f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\frac{1}{\sqrt{2}} - \sqrt{2}}{\sqrt{2}} \right)$$

$$f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \tan^{-1} \left(-\frac{1}{2} \right)$$

11. If $\int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x-\theta)}} dx = A\sqrt{\cos \theta \tan x - \sin \theta} + B\sqrt{\cos \theta - \sin \theta \cot x} + C$, where C is the integration

constant, then AB is equal to :

यदि $\int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x-\theta)}} dx = A\sqrt{\cos \theta \tan x - \sin \theta} + B\sqrt{\cos \theta - \sin \theta \cot x} + C$ है, जहाँ C समाकलन अचर

है, तो AB बराबर है :

- (A) 4 sec θ (B) 4 cosec (2 θ) (C) 2 sec θ (D) 8 cosec (2 θ)

Ans. D

Sol.
$$I = \int \frac{\left(\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x \right) dx}{\sqrt{\sin^3 x \cos^3 x \sin(x - \theta)}}$$

$$I = \int \frac{\sec^{\frac{3}{2}} x dx}{\sqrt{\sin(x - \theta)}} + \int \frac{\operatorname{cosec}^{\frac{3}{2}} x dx}{\sqrt{\sin(x - \theta)}}$$

$$I = \int \frac{\sec^2 x dx}{\sqrt{\sec x \sin(x - \theta)}} + \int \frac{\operatorname{cosec}^2 x dx}{\sqrt{\operatorname{cosec} x \sin(x - \theta)}}$$

$$I = \int \frac{\sec^2 x dx}{\sqrt{\tan x \cos \theta - \sin \theta}} + \int \frac{\operatorname{cosec}^2 x dx}{\sqrt{\cos \theta - \cot x \sin \theta}} = I_1 + I_2$$

$$I_1 = \int \frac{\sec^2 x dx}{\sqrt{\tan x \cos \theta - \sin \theta}}$$

Let $\tan x \cos \theta - \sin \theta = t^2$

$$\Rightarrow \sec^2 x dx = \frac{2tdt}{\cos \theta}$$

$$I_1 = \int \frac{2tdt}{\frac{\cos \theta}{t}} = \frac{2t}{\cos \theta} + C_1$$

$$= \frac{2}{\cos \theta} \sqrt{\tan x \cot \theta - \sin \theta} + C_1$$

$$I_2 = \frac{2}{\sin \theta} \sqrt{\cos \theta - \sin \theta \cot x} + C_2$$

$$\Rightarrow I_1 + I_2 \Rightarrow B = \frac{2}{\sin \theta}, A = \frac{2}{\cos \theta}$$

$$\Rightarrow AB = \frac{4}{\sin \theta \cos \theta} = 8 \operatorname{cosec} 2\theta$$

12. Let $I(x) = \int \frac{x^2 (x \sec^2 x + \tan x)}{(x \tan x + 1)^2} dx$. If $I(0) = 0$, then $I\left(\frac{\pi}{4}\right)$ is equal to :

माना $I(x) = \int \frac{x^2 (x \sec^2 x + \tan x)}{(x \tan x + 1)^2} dx$ है। यदि $I(0) = 0$ है, तब $I\left(\frac{\pi}{4}\right)$ का मान है :

$$(A) \log_e \frac{(\pi+4)^2}{16} + \frac{\pi^2}{4(\pi+4)}$$

$$(B) \log_e \frac{(\pi+4)^2}{32} - \frac{\pi^2}{4(\pi+4)}$$

$$(C) \log_e \frac{(\pi+4)^2}{32} + \frac{\pi^2}{4(\pi+4)}$$

$$(D) \log_e \frac{(\pi+4)^2}{16} - \frac{\pi^2}{4(\pi+4)}$$

Ans. B

Sol.
$$\int x^2 \left(\frac{x \sec^2 x + \tan x}{(x \tan x + 1)^2} \right) dx$$

$$= \frac{x^2}{(x \tan x + 1)} + \int \frac{2x}{x \tan x + 1} dx$$

$$I = 2 \int \frac{x}{x \tan x + 1} dx$$

$$= 2 \int \frac{x \cos x}{x \sin x + \cos x} dx$$

Let $x \sin x + \cos x = t$

$$(x \cos x + \sin x - \sin x) dx = dt$$

$$= 2 \int \frac{dt}{t} = 2 \log t + c$$

$$= 2 \log |x \sin x + \cos x| + c$$

$$\therefore \int \frac{x^2 (x \sec^2 x + \tan x)}{(x \tan x + 1)^2} dx = \frac{-x^2}{x \tan x + 1} + 2 \log |x \sin x + \cos x| + c$$

$$I(0) = 0 \quad \Rightarrow c = 0$$

$$I\left(\frac{\pi}{4}\right) = \frac{-\pi}{\frac{\pi}{4} \times 1 + 1} + 2 \log \left| \frac{1}{\sqrt{2}} \left(\frac{\pi}{4} \right) + 1 \right|$$

$$= \log_e \frac{(\pi+4)^2}{32} - \frac{\pi^2}{4(\pi+4)}$$

13. If $I(x) = \int e^{\sin^2 x} (\cos x \sin 2x - \sin x) dx$ and $I(0) = 1$, then $I\left(\frac{\pi}{3}\right)$ is equal to :

यदि $I(x) = \int e^{\sin^2 x} (\cos x \sin 2x - \sin x) dx$ तथा $I(0) = 1$ हैं, तो $I\left(\frac{\pi}{3}\right)$ बराबर है :

(A) $e^{\frac{3}{4}}$

(B) $-\frac{1}{2}e^{\frac{3}{4}}$

(C) $-e^{\frac{3}{4}}$

(D) $\frac{1}{2}e^{\frac{3}{4}}$

Ans. D

Sol. $\therefore I = \int e^{f(x)} (g'(x) + f'(x) \cdot g(x)) dx = e^{f(x)} \cdot g(x)$

$\therefore I = e^{\sin^2 x} \cdot \cos x + c$

but $I(0) = 1 \Rightarrow c = 0$

$I = e^{\sin^2 x} \cos x$

$I\left(\frac{\pi}{3}\right) = \frac{1}{2}e^{\frac{3}{4}}$

14. For $\alpha, \beta, \gamma, \delta \in \mathbb{N}$, if $\int \left(\left(\frac{x}{e} \right)^{2x} + \left(\frac{e}{x} \right)^{2x} \right) \log_e x \, dx = \frac{1}{\alpha} \left(\frac{x}{e} \right)^{\beta x} - \frac{1}{\gamma} \left(\frac{e}{x} \right)^{\delta x} + C$, where $e = \sum_{n=0}^{\infty} \frac{1}{n!}$ and C is constant of integration, then $\alpha + 2\beta + 3\gamma - 4\delta$ is equal to :

$\alpha, \beta, \gamma, \delta \in \mathbb{N}$ के लिए, यदि $\int \left(\left(\frac{x}{e} \right)^{2x} + \left(\frac{e}{x} \right)^{2x} \right) \log_e x \, dx = \frac{1}{\alpha} \left(\frac{x}{e} \right)^{\beta x} - \frac{1}{\gamma} \left(\frac{e}{x} \right)^{\delta x} + C$ है, जहाँ $e = \sum_{n=0}^{\infty} \frac{1}{n!}$ तथा C

समाकलन अचर है, तो $\alpha + 2\beta + 3\gamma - 4\delta$ बराबर है :

(A) -4

(B) 1

(C) 4

(D) -8

Ans. C

Sol. Let $\left(\frac{x}{e} \right)^{2x} = t$

$2x(\ln x - 1) = \ln t$

$\left(2(\ln x - 1) + 2x \times \frac{1}{x} \right) dx = \frac{1}{t} dt$

$\ln x dx = \frac{1}{2t} dt$

$I = \int \left(t + \frac{1}{t} \right) \times \frac{1}{2t} dt = \frac{1}{2} \int \left(1 + \frac{1}{t^2} \right) dt$

$= \frac{1}{2} \left(t - \frac{1}{t} \right) + C$

$$= \frac{1}{2} \left(\frac{x}{e} \right)^{2x} - \frac{1}{2} \left(\frac{e}{x} \right)^{2x} + C$$

$$\Rightarrow \alpha = 2, \beta = 2, \gamma = 2, \delta = 2$$

$$\therefore \alpha + 2\beta + 3\gamma - 4\delta = 2 + 4 + 6 - 8 = 4$$

15. If $\int \frac{(\sqrt{1+x^2} + x)^{10}}{(\sqrt{1+x^2} - x)^9} dx = \frac{1}{m} \left((\sqrt{1+x^2} + x)^n (n\sqrt{1+x^2} - x) \right) + C$ where C is the constant of integration and m,

$n \in \mathbb{N}$, then $m + n$ is equal to _____.

यदि $\int \frac{(\sqrt{1+x^2} + x)^{10}}{(\sqrt{1+x^2} - x)^9} dx = \frac{1}{m} \left((\sqrt{1+x^2} + x)^n (n\sqrt{1+x^2} - x) \right) + C$ है, जहाँ C समाकलन अचर है और $m, n \in \mathbb{N}$ हैं, तो

$m + n$ बराबर है _____.

Ans. 379

Sol. $\int \frac{(\sqrt{1+x^2} + x)^{10}}{(\sqrt{1+x^2} - x)^9} dx$

Put $x = \tan \theta$

$$\int \frac{(\sec \theta + \tan \theta)^{10}}{(\sec \theta - \tan \theta)^9} (\sec^2 \theta d\theta)$$

$$\int (\sec \theta + \tan \theta)^{19} (\sec^2 \theta) d\theta$$

Put $\sec \theta + \tan \theta = z$; $\sec \theta (\sec \theta + \tan \theta) d\theta = dz$

$$\sec \theta - \tan \theta = \frac{1}{z}$$

$$\sec \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$\Rightarrow \int z^{18} \left(\frac{1}{2} \left(z + \frac{1}{z} \right) \right) dz$$

$$\Rightarrow \frac{1}{2} \int (z^{19} + z^{17}) dz$$

$$\begin{aligned}
 &\Rightarrow \frac{1}{2} \left(\frac{z^{20}}{20} + \frac{z^{18}}{18} \right) + c \\
 &= \frac{1}{2} \left(\frac{(\sec \theta + \tan \theta)^{20}}{20} + \frac{(\sec \theta + \tan \theta)^{18}}{18} \right) + c \\
 &= \frac{1}{2} \times \frac{(\sqrt{1+x^2} + x)^{18}}{2} \left\{ \frac{(\sqrt{1+x^2} + x)^2}{10} + \frac{1}{9} \right\} + c \\
 &\Rightarrow \frac{1}{4} (\sqrt{1+x^2} + x)^{19} \left\{ \frac{\sqrt{1+x^2} + x}{10} + \frac{1}{9} (\sqrt{1+x^2} - x) \right\} + c \\
 &\Rightarrow \frac{1}{360} (\sqrt{1+x^2} + x)^{19} \{ 19(\sqrt{1+x^2}) - x \} + c \\
 &\Rightarrow m = 360, n = 19
 \end{aligned}$$

16. If $\int \left(\frac{1}{x} + \frac{1}{x^3} \right) \left(\sqrt[23]{3x^{-24} + x^{-26}} \right) dx = -\frac{\alpha}{3(\alpha+1)} (3x^\beta + x^\gamma)^{\frac{\alpha+1}{\alpha}} + C, x > 0, (\alpha, \beta, \gamma \in \mathbb{Z})$, where C is the constant of integration, then $\alpha + \beta + \gamma$ is equal to _____.

यदि $\int \left(\frac{1}{x} + \frac{1}{x^3} \right) \left(\sqrt[23]{3x^{-24} + x^{-26}} \right) dx = -\frac{\alpha}{3(\alpha+1)} (3x^\beta + x^\gamma)^{\frac{\alpha+1}{\alpha}} + C, x > 0, (\alpha, \beta, \gamma \in \mathbb{Z})$, जहाँ C समाकलन अचर है, तो $\alpha + \beta + \gamma$ बराबर है _____.

Ans. 19

Sol.
$$I = \int \left(\frac{1}{x} + \frac{1}{x^3} \right) \left(\frac{3}{x^{24}} + \frac{1}{x^{26}} \right)^{\frac{1}{23}} dx$$

$$I = \int \left(\frac{1}{x^2} + \frac{1}{x^4} \right) \left(\frac{3}{x} + \frac{1}{x^3} \right)^{\frac{1}{23}} dx$$

Let $\frac{3}{x} + \frac{1}{x^3} = t$

$$\Rightarrow \left(-\frac{3}{x^2} - \frac{3}{x^4} \right) dx = dt$$

$$\text{so, } I = -\frac{1}{3} \int t^{\frac{1}{23}} dt$$

$$= -\frac{1}{3} \cdot \frac{23}{24} t^{\frac{24}{23}} + c$$

$$= -\frac{1}{3} \cdot \frac{23}{24} \left(\frac{3}{x} + \frac{1}{x^3} \right)^{\frac{24}{23}} + c$$

$$= -\frac{1}{3} \cdot \frac{23}{24} (3x^{-1} + x^{-3})^{\frac{24}{23}} + c$$

$$\text{comparing} \quad \alpha = 23, \beta = -1, \gamma = -3$$

$$\Rightarrow \alpha + \beta + \gamma = 19$$

17. If $\int \frac{2x^2 + 5x + 9}{\sqrt{x^2 + x + 1}} dx = x\sqrt{x^2 + x + 1} + \alpha\sqrt{x^2 + x + 1} + \beta \log_e \left| x + \frac{1}{2} + \sqrt{x^2 + x + 1} \right| + C$, where C is the constant of integration, then $\alpha + 2\beta$ is equal to _____.

यदि $\int \frac{2x^2 + 5x + 9}{\sqrt{x^2 + x + 1}} dx = x\sqrt{x^2 + x + 1} + \alpha\sqrt{x^2 + x + 1} + \beta \log_e \left| x + \frac{1}{2} + \sqrt{x^2 + x + 1} \right| + C$ है, जहाँ C समाकलन अचर है, तो $\alpha + 2\beta$ बराबर है _____.

Ans. 16

Sol. Diffⁿ both sides;

$$\frac{2x^2 + 5x + 9}{\sqrt{x^2 + x + 1}} = x \left(\frac{2x + 1}{2\sqrt{x^2 + x + 1}} \right) + \sqrt{x^2 + x + 1} + \alpha \left(\frac{2x + 1}{2\sqrt{x^2 + x + 1}} \right) + \beta \left(\frac{1 + \frac{2x + 1}{2\sqrt{x^2 + x + 1}}}{x + \frac{1}{2} + \sqrt{x^2 + x + 1}} \right)$$

$$\Rightarrow \frac{2x^2 + 5x + 9}{\sqrt{x^2 + x + 1}} = \frac{2x^2 + x + 2x^2 + 2x + 2 + 2\alpha x + \alpha}{2\sqrt{x^2 + x + 1}} + \beta \left(\frac{2\sqrt{x^2 + x + 1} + 2x + 1}{2\sqrt{x^2 + x + 1}} \right) \left(\frac{2}{2x + 1 + 2\sqrt{x^2 + x + 1}} \right)$$

$$\Rightarrow \frac{2x^2 + 5x + 9}{\sqrt{x^2 + x + 1}} = \frac{4x^2 + (3 + 2\alpha)x + 2 + \alpha + 2\beta}{2\sqrt{x^2 + x + 1}}$$

$$\Rightarrow 5 = \frac{3 + 2\alpha}{2} \text{ and } 9 = \frac{2 + \frac{7}{2} + 2\beta}{2}$$

$$\alpha = \frac{7}{2} \quad 16 - \frac{7}{2} = 2\beta \quad \Rightarrow \beta = \frac{25}{4}$$

$$\Rightarrow \alpha + 2\beta = \frac{7}{2} + \frac{25}{2} = 16$$

18. If $\int \cos ec^5 x dx = \alpha \cot x \cos ecx \left(\cos ec^2 x + \frac{3}{2} \right) + \beta \log_e \left| \tan \frac{x}{2} \right| + C$ where $\alpha, \beta \in \mathbb{R}$ and C is the constant of integration, then the value of $8(\alpha + \beta)$ equals _____.

यदि $\int \cos ec^5 x dx = \alpha \cot x \cos ecx \left(\cos ec^2 x + \frac{3}{2} \right) + \beta \log_e \left| \tan \frac{x}{2} \right| + C$ है, जहाँ $\alpha, \beta \in \mathbb{R}$ तथा C समाकल अचर है, तो $8(\alpha + \beta)$ का मान बराबर है _____

Ans. 1

Sol. $\int \cos ec^3 x \cdot \cos ec^2 x dx = I$

By applying integration by parts

$$I = -\cot x \cos ec^3 x + \int \cot x (-3 \cos ec^2 x \cot x \cos ecx) dx$$

$$I = -\cot x \cos ec^3 x - 3I - 3 \int \cos ec^3 x (\cos ec^2 x - 1) dx$$

$$I = -\cot x \cos ec^3 x - 3I + 3 \int \cos ec^3 x dx$$

Let

$$I_1 = \int \cos ec^3 x dx = -\cos ecx \cot x - \int \cot^2 x \cos ecx dx$$

$$I_1 = -\cos ecx \cot x - \int (\cos ec^2 x - 1) \cos ecx dx$$

$$2I_1 = -\cos ecx \cot x + \ell n \left| \tan \frac{x}{2} \right|$$

$$I_1 = -\frac{1}{2} \cos ecx \cot x + \frac{1}{2} \ell n \left| \tan \frac{x}{2} \right|$$

$$4I = -\cot x \cos ec^3 x - \frac{3}{2} \cos ecx \cot x + \frac{3}{2} \ell n \left| \tan \frac{x}{2} \right| + 4c$$

$$I = -\frac{1}{4} \cos ecx \cot x \left(\cos ec^2 x + \frac{3}{2} \right) + \frac{3}{2} \ell n \left| \tan \frac{x}{2} \right| + c$$

$$\therefore \alpha = \frac{-1}{4}, \beta = \frac{3}{8} \rightarrow 8(\alpha + \beta) = 1$$

19. Let $I(x) = \int \sqrt{\frac{x+7}{x}} dx$ and $I(9) = 12 + 7 \log_e 7$. If $I(1) = \alpha + 7 \log_e (1 + 2\sqrt{2})$, α^4 is equal to _____.

माना $I(x) = \int \sqrt{\frac{x+7}{x}} dx$ तथा $I(9) = 12 + 7 \log_e 7$ हैं। यदि $I(1) = \alpha + 7 \log_e (1 + 2\sqrt{2})$ है, तो α^4 बराबर है _____

Ans. 64

Sol. $I(x) = \int \sqrt{\frac{x+7}{x}} dx$

$$= \int \frac{\sqrt{(\sqrt{x})^2 + (\sqrt{7})^2}}{\sqrt{x}} dx$$

$$\text{Let } \sqrt{x} = t \therefore \frac{1}{2\sqrt{x}} dx = dt$$

$$= 2 \int \sqrt{t^2 + (\sqrt{7})^2} dt = 2 \left(\frac{t}{2} \sqrt{t^2 + 7} + \frac{7}{2} \ln(t + \sqrt{t^2 + 7}) \right) + c$$

$$I(x) = \sqrt{x} \sqrt{x+7} + 7 \ln(\sqrt{x} + \sqrt{x+7}) + c$$

$$I(9) = 12 + 7 \ln 7 = 12 + 7 \ln(3+4) + c$$

$$\Rightarrow c = 0$$

$$\therefore I(x) = \sqrt{x} \sqrt{x+7} + 7 \ln(\sqrt{x} + \sqrt{x+7})$$

$$\text{at } x = 1$$

$$I(1) = \sqrt{8} + 7 \ln(1 + \sqrt{8}) = 7 \ln(1 + 2\sqrt{2}) + \alpha$$

$$\alpha = \sqrt{8}$$

$$\alpha^4 = 64$$

20. Let $f(x) = \int \frac{dx}{(3+4x^2)\sqrt{4-3x^2}}$, $|x| < \frac{2}{\sqrt{3}}$. If $f(0) = 0$ and $f(1) = \frac{1}{\alpha\beta} \tan^{-1}\left(\frac{\alpha}{\beta}\right)$, $\alpha, \beta > 0$, then $\alpha^2 + \beta^2$ is equal to _____.

यदि $f(x) = \int \frac{dx}{(3+4x^2)\sqrt{4-3x^2}}$, $|x| < \frac{2}{\sqrt{3}}$, $f(0) = 0$ तथा $f(1) = \frac{1}{\alpha\beta} \tan^{-1}\left(\frac{\alpha}{\beta}\right)$, $\alpha, \beta > 0$ हो, तो $\alpha^2 + \beta^2$ का मान

होगा _____

Ans. 28

Sol. $f(x) = \int \frac{dx}{(3+4x^2)\sqrt{4-3x^2}}$

Let $x = \frac{1}{t}$

$dx = -\frac{1}{t^2} dt$

$= \int \frac{-\frac{1}{t^2} dt}{\left(3 + \frac{4}{t^2}\right)\sqrt{4 - \frac{3}{t^2}}}$

$= \int \frac{-tdt}{(3t^2 + 4)\sqrt{4t^2 - 3}}$

Now, let $4t^2 - 3 = p^2$

$4t dt = p dp$

$= \int \frac{-\frac{p}{4} dp}{\left(\frac{3p^2 + 25}{4}\right)p} = -\int \frac{dp}{3p^2 + 25}$

$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}p}{5}\right) + C$

$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3} \cdot \sqrt{4t^2 - 3}}{5}\right) + C$

$= -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3} \cdot \sqrt{\frac{4}{x^2} - 3}}{5}\right) + C$

$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3} \cdot \sqrt{4 - 3x^2}}{5|x|}\right) + C$

$x = 0, y = 0$

$$0 = -\frac{1}{5\sqrt{3}} \tan^{-1}(\infty) + C$$

$$C = \frac{1}{5\sqrt{3}} \times \frac{\pi}{2}$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1} \frac{\sqrt{3}\sqrt{4-3x^2}}{5x} + \frac{\pi}{2(5\sqrt{3})}$$

$$f(1) = -\frac{1}{5\sqrt{3}} \tan^{-1} \frac{\sqrt{3}}{5} + \frac{\pi}{2(5\sqrt{3})}$$

$$= \frac{1}{5\sqrt{3}} \left(\frac{\pi}{2} - \tan^{-1} \frac{\sqrt{3}}{5} \right)$$

$$= \frac{1}{5\sqrt{3}} \cot^{-1} \frac{\sqrt{3}}{5}$$

$$= \frac{1}{5\sqrt{3}} \tan^{-1} \frac{5}{\sqrt{3}}$$

$$\text{Now, } \alpha = 5, \beta = \sqrt{3}$$

$$\alpha^2 + \beta^2 = 25 + 3 = 28$$